

CLICK-A-THON 2026 · ATLYS TRACK

From feature spec to insight — automatically.

An agentic analytics system on ClickHouse that instruments, analyzes, and explains.

3 traced agents

ClickHouse Cloud

LibreChat + Subagents

Vector + OTel

Langfuse + ClickStack

THE PROBLEM

Every new feature restarts a slow, manual loop

- **Instrument** — someone decides what to track and hand-writes a tracking PRD.
- **Schema** — engineers design & create ClickHouse tables weeks later.
- **Analyze** — analysts write queries, chase segments, build charts.
- **Explain** — by the time insights land, the team has moved on.

Context is lost across every handoff — and a **surprise 6th feature spec** drops in the final hours. Systems tuned to the 5 known specs show their seams.

OUR SOLUTION

Three agents that collapse the loop into one traced run

AGENT 1

Instrumentation

Reads any feature spec, designs optimal ClickHouse DDL (types, ORDER BY, partitioning, MVs), validates + smoke-tests on Cloud, and pushes the schema to GitHub.

AGENT 3

Context Keeper

Maintains a living OKF context bundle. On every new table it bumps the version, logs a changelog diff, and surfaces contradictions in the imperfect base context.

AGENT 2

Analytics

Reads the fresh context, pushes aggregation into ClickHouse (never raw rows), and writes PM-ready insights with the *why* and confidence scores.

Orchestrated by **LibreChat Subagents** — Instrumentation is the parent: **spec** → **schema** → **context** → **insight**, one Langfuse trace end to end.

All three agents run inside LibreChat

Not a bespoke framework — each agent is a configured **LibreChat Agent** (system prompt + skills + tools), chained through native **Subagents**. Agents have no shell: every DB, git, and file action goes through MCP servers.

Subagent chain

Instrumentation is the parent and invokes Context, then Analytics, as isolated subagents — keeping conversation control and combining the result.

MCP tools only

DDL + smoke-tests, read-only SELECTs, git push, and context files are all exposed as MCP servers — safe, auditable tool boundaries.

Conversational UI

The same LibreChat surface is the PM's chat with the Analytics agent for on-demand, ad-hoc questions — every turn traced.

Native tracing

LibreChat's Langfuse integration traces each agent + the full chain; models served as opus-4.x via a LiteLLM gateway.

clickhouse_write_tools

clickhouse-cloud (read-only)

clickhouse_git_write

filesystem

Pipeline result: Promo / Coupon at Checkout

Pipeline Result: `unseen_data` (Promo / Coupon at Checkout)

5 insights computed from live ClickHouse data | 5,363 events, 2,100 users | Jun 8–28 2026 | Context v7

Insight	Metric	Key Value	Confidence
I1 — Apply rate is 27.6%, but 31.6% of code entries are rejected	M1: Coupon Apply Rate	580 applied / 2,100 shown	High (0.90)
I2 — EXPIRED5 code generates 56% of all rejections — retire it	M2: Reject Mix	149/268 from EXPIRED5; #1 reason: <code>min_cart_not_met</code> (30%)	High (0.85)
I3 — Coupon users convert at 1.54× the baseline (63.1% vs 40.9%)	M3: Conversion Lift	Android strongest at 1.77×; iOS weakest at 1.39×	Medium (0.75)
I4 — SUMMER20 costs ₹1,055/event at 20% discount; total margin erosion ₹813K	M4: Margin Cost	SUMMER20 > ATLYS15 > FIRST10 > WELCOME > FREESHIP(₹0)	High (0.85)
I5 — Android has strongest lift but highest actionable reject rate (34%); Thailand is the most margin-erosive destination (8.7%)	M5: Segment Performance	iOS leads volume; Desktop worst reject rate (40%)	High (0.80)

Top action items for PM

- 1. **Kill or replace EXPIRED5** — it's a dead code generating half of all rejections and frustrating ~150 users
- 2. **Show min-cart threshold before code entry** — `min_cart_not_met` is the #1 reject reason and entirely preventable with UX
- 3. **Cap SUMMER20 or convert to flat discount** — at 20% it's the most expensive code and its ₹339K total discount dwarfs others
- 4. **Fix Desktop coupon field prominence** — 40% reject rate suggests users enter codes speculatively

Autonomous run — 5 insights computed from live ClickHouse (5,363 events · 2,100 users), each with a metric, key value, confidence score, and PM action items.
Context v7.

INSIGHT QUALITY · THE "WHY", NOT JUST THE "WHAT"

Insights a PM would act on

- 1. **M1 (Apply Rate):** 27.6% of users who see the coupon field successfully apply — with a 40.4% entry rate and 31.6% rejection rate among those who attempt
- 2. **M2 (Reject Mix):** EXPIRED5 code drives 56% of all rejections (149/268). Top reason across all codes is `min_cart_not_met` (29.9%), followed by `already_used` (28.0%), `expired` (22.4%), `invalid_code` (19.8%)
- 3. **M3 (Conversion Lift):** Coupon users convert at 63.1% vs baseline 40.9% — a **1.54× lift** (+22.2pp). Android shows the strongest lift (67.8% vs 38.3%)
- 4. **M4 (Margin Cost):** SUMMER20 is the costliest coupon at ~20% discount rate (₹338,623 total). FREESHIP costs ₹0 in discount. Total coupon discount: ₹813,317 on ₹6,344,375 of couponed cart value (12.8% margin cost)
- 5. **M5 (Segment Performance):** iOS leads in volume (874 users) and apply rate (29.1%); Android shows strongest conversion lift. GB and AU show highest apply rates (~31.5%); Turkey trails at 22.6%. Thailand has the highest margin cost per destination at 8.7%.

Apply rate, reject mix, conversion lift, margin cost, and segment performance — cross-cut by device, geo, and destination.

1.54×

coupon conversion lift vs baseline

56%

of rejections from one dead code
(EXPIRED5)

₹813K

total margin erosion identified

40%

desktop reject rate — worst segment

The Context Agent catches what's stale & contradictory

Context contradictions surfaced

- `coupon-reject-designed-not-observed.md` claimed zero `coupon_rejected` rows — **stale**: 268 rows now present in populated data
- M3 computed via base-table self-join per `coupon-conversion-lift-single-table-gap.md` (funnel agg cannot express per-user sequencing)
- K6 (SUMMER20 campaign) confirmed: code is present and active with the highest per-event discount (₹1,055 avg)

On the new table landing, the agent re-checks the base context and flags a claim that is now stale — proving Analytics reasons from just-updated context.

Provable freshness

Versioned Markdown bundle + newest-first changelog — every update is a diff a judge can read.

Compute in ClickHouse

LLM only narrates aggregates — never streams raw rows. Token-safe by design.

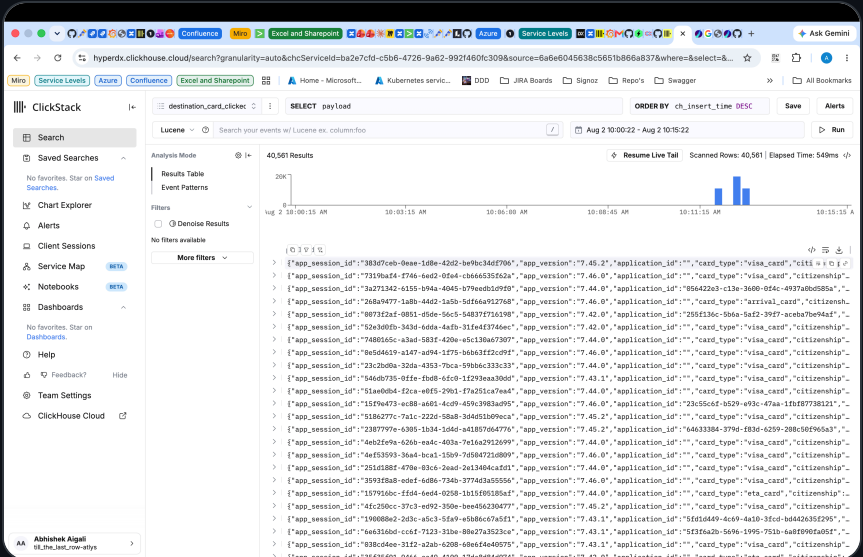
No trace, no credit

Every agent step is a named Langfuse span; the full chain is one trace.

Vector + OpenTelemetry Collector → ClickHouse

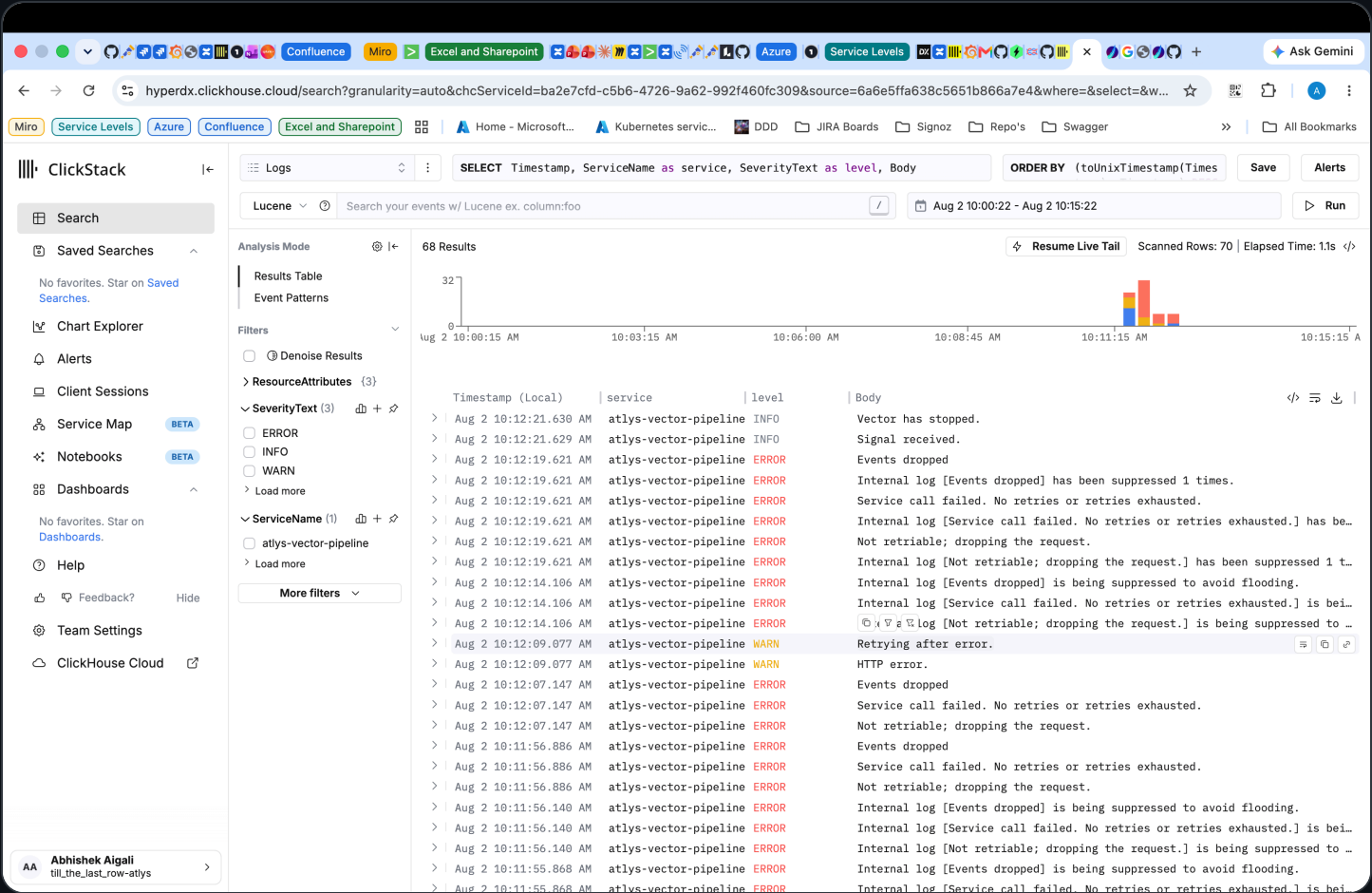
A containerized **Vector** + **OTel Collector** pipeline transforms events and streams them into ClickHouse Cloud — decoupled from the agent flow.

☐	▼	●	vector-pipeline	-	-	-	0.59%	15 seconds			
☐		●	atlys-otelcol	592c5402200d	otel/opentelemetry-collector	24224:24224	0.38%	15 seconds			
					Show all ports (4)						
☐		●	atlys-vector	e056096ca521	vector-pipeline		0.21%	15 seconds			



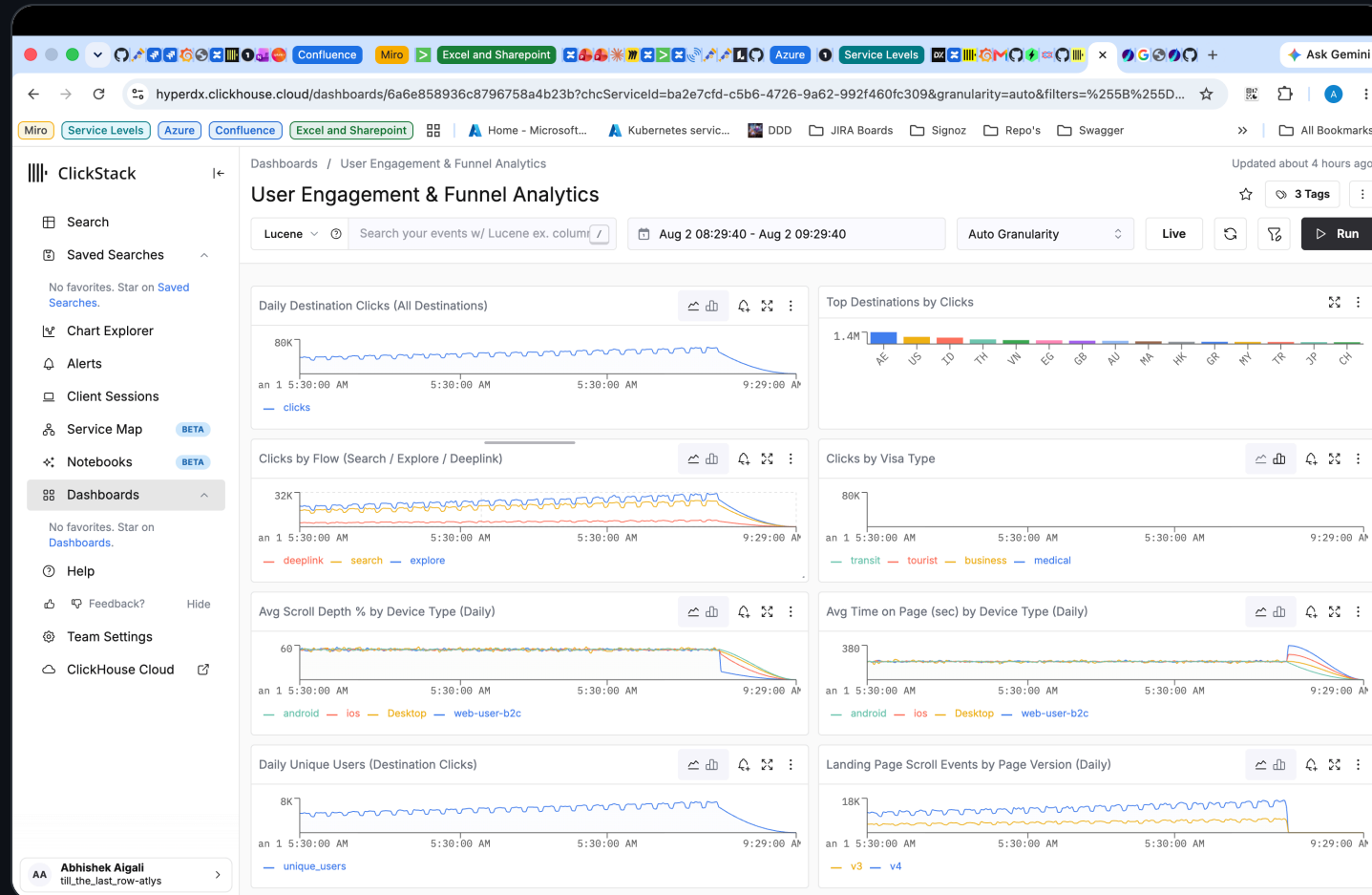
Live services: **atlys-vector** + **atlys-otelcol** feeding ClickHouse — 40,561 events landed and queryable within seconds.

Full platform observability on ClickStack



Pipeline logs, severities, and service health for **atlys-vector-pipeline** in HyperDX / ClickStack — complementing Langfuse (LLM traces) with system-level observability.

User Engagement & Funnel Analytics dashboard



Destination clicks, flow & visa-type breakdowns, scroll depth and time-on-page by device, and daily unique users — all served live from ClickHouse in ClickStack.

Schema changes over time — with agent insights

127.0.0.1:8777

service Levels Azure Confluence Excel and Sharepoint Home - Microsoft... Kubernetes servic... DDD JIRA Boards Signoz Repo's Swagger

Atlys - Schema Changes Over Time

7169317168

git history + .sql headers + .metrics.json + .insights.json + context_docs/log.md

SCHEMAS TABLES MAT. VIEWS METRICS INSIGHTS SCHEMA COMMITS CONTEXT VERSIONS

LINE SCHEMA COMMIT CONTEXT VERSION

2026-08-07

Context v8 — schema-change: 02_group_family (Group / Family Applications) landed live in ClickHouse (base + daily agg + MV)

+11 added -3 updated

show diff (14)

source: 'Atlys/schemas/02_group_family.sql', 'Atlys/schemas/02_group_family.metrics.json' (4 metrics, all confirmed_by_user=true), 'specs/02_group_family/spec.md', live 'atlys' schema (system.tables/columns/data_skipping_indices + create_table_query for group_family, group_family_daily, group_family_daily_mv)

2026-08-07

Context v7 — schema-change: unseen_data (Promo / Coupon at Checkout, sealed 6th spec) landed live in ClickHouse (base + 2 aggs + 2 MVs)

+8 added -1 updated

show diff (9)

source: 'specs/unseen_data/spec.md', 'specs/unseen_data/events.ndjson', live 'atlys' schema (system.tables/columns/data_skipping_indices + create_table_query for promo_coupon_checkout, coupon_funnel_daily_agg, coupon_discount_daily_agg, coupon_funnel_daily_mv, coupon_discount_daily_mv). NOTE: Atlys/schemas/unseen_data.sql and .metrics.json were NOT accessible from allowed dirs (/app/context_docs, /app/specs) - grounded against Live ClickHouse + spec instead.

2026-08-06

Context v6 — schema-change: 11_document_uploaded landed live in ClickHouse (base + daily agg + MV)

+7 added -6 updated

show diff (13)

source: live 'atlys' schema (system.tables/columns/data_skipping_indices + create_table_query); Atlys/schemas/11_document_uploaded.sql + .metrics.json (5 metrics, all confirmed_by_user=false); specs/11_document_uploaded/spec.md

SCHEMAS

01_express_checkout

3 tables 2 MV 0 idx 0 metrics 0 insights 2 rev

07_landing_page_scrolled

2 tables 1 MV 7 idx 5 metrics 0 insights 1 rev

08_destination_card_clicked

2 tables 1 MV 6 idx 6 metrics 3 insights 1 rev

09_auth_completed

2 tables 1 MV 5 idx 5 metrics 0 insights 1 rev

10_application_started

2 tables 1 MV 6 idx 5 metrics 0 insights 1 rev

11_document_uploaded

2 tables 1 MV 7 idx 5 metrics 4 insights 1 rev

unseen_data

3 tables 2 MV 4 idx 5 metrics 0 insights 1 rev

DDL OBJECTS

atlys.promo_coupon_checkout atlys.coupon_funnel_daily_agg atlys.coupon_discount_daily_agg atlys.coupon_funnel_daily_mv atlys.coupon_discount_daily_mv atlys

ORDER BY

payload.event, payload.application_id, payload.device_type, payload.geoup_country_code, payload.timestamp

PARTITION BY

127.0.0.1:8777

Miro service Levels Azure Confluence Excel and Sharepoint Home - Microsoft... Kubernetes servic... DDD JIRA Boards Signoz Repo's Swagger

Atlys - Schema Changes Over Time

git history + .sql headers + .metrics.json + .insights.json + context_docs/log.md

7169317168

AGENT-GENERATED INSIGHTS - with confidence scores

7 insights 3 High 4 Medium known issues: K1 K2 K3 K7

11_document_uploaded High

Data quality gap: os field is empty for 18% of Android uploads (D3 contradiction confirmed), and M5 conversion metric is blocked by missing purchase_completed table

98%

CONFIDENCE

D3 is directly observable in the data. M5 gap is a schema-level fact. K3 non-finding is limited by the citizenship granularity available (bucketed, not per-country for smaller populations).

EVIDENCE

D3 (android-os=null): 1,188 of 6,541 Android uploads (18.2%) have os="". Non-Latin passport MRZ (K3): citizenship-based retry analysis shows no significant pattern — 'bd' (Bangladesh, non-Latin) avg_retries=0.48 vs 'in' (India, Latin) 0.45 vs 'de' (Germany, Latin) 0.53 — differences are within noise. M5 (doc-volume-vs-payment-conversion): purchase_completed table does not exist in the database; pay_now_clicked (spec 12) is not live. Cross-table funnel analysis is currently impossible.

NEXT STEP

Fix the Android SDK to populate the os field reliably. Prioritize specs 12 (pay_now_clicked) and 13 (purchase_completed) to unblock M5 and full funnel analysis from document upload to purchase.

K3 doc-volume-vs-payment-conversion

11_document_uploaded High

Android passport upload failures are surging 5x since Jan, consistent with K2 passport scan model update

85%

CONFIDENCE

Strong temporal correlation with K2, clear Android-only signal across 6,541 uploads, iOS control group stable. Hedged because we cannot prove causation without A/B or rollback data.

EVIDENCE

Android threshold-crossing rate (M2): is_crossed_failed_attempt_threshold=1 / total uploads: Jan 6.6% → Mar 7.9% → Apr 12.1% → May 23.1% → Jun 33.8%. iOS stayed flat: Jan 8.2% → Jun 10.2%. Android n=6,541 uploads (32% of total); iOS n=8,479 (41%). The inflection begins in April, coinciding with K2 (on-device passport model update, early April). Both auto and manual scan modes on Android show the same acceleration (Jun: auto 34.1%, manual 31.7%), indicating the issue is scan-mode-agnostic.

WHY

The timing and Android-only scope are consistent with K2 (passport scan model update deployed to Android in early April 2026). The model change appears to have introduced progressive degradation — not a one-time step change — suggesting a model-confidence decay or data-pipeline feedback loop. The fact that both auto and manual scan modes are equally affected rules out an auto-scan-specific regression.

NEXT STEP

08_destination_card_clicked High

Guest-browse card clicks convert to application_started ~38% worse than logged-in clicks.

82%

CONFIDENCE

Large sample, stable week-over-week, metric defined in context.

EVIDENCE

guest 6.1% vs logged-in 9.8% (application_started / destination_card_clicked), 2026-05-01+08-01, n=41,203 clicks.

WHY

Guests hit the auth wall before they can start an application; consistent with the funnel's known auth drop, not a card/page defect.

NEXT STEP

A/B a lightweight guest checkout that defers auth until after application_started.

guest_browse_conversion guest_browse_click_share Langfuse trace

11_document_uploaded Medium

11.2% of all document uploads hit the failure threshold — and camera capture (70% of volume) drives the bulk of friction

70%

CONFIDENCE

Solid sample sizes across all capture modes (camera n=14,305; gallery n=4,443; QR n=1,698). Medium rather than High because all doc_type = passport_front — we cannot assess whether other document types would show different patterns.

EVIDENCE

Overall threshold-crossing rate (M2): 11.2% (2,299/20,446). Retry histogram (M1): 70.1% success on first try (retry_count=0), 18.2% retry once, 7.9% retry twice, 3.9% retry 3+ times (threshold). By capture_mode: camera 70.0% share / 11.3% threshold rate; gallery 21.7% / 11.3%; QR 8.3% / 10.6%. Avg retries nearly identical across capture modes (~0.45). The monthly trend shows overall threshold-crossing rising from 7.8% (Jan) to 17.7% (Jun), driven almost entirely by the Android surge (Insight I1).

WHY

Camera dominates volume because most mobile users capture live passport photos. The similar threshold-crossing rates across capture modes suggest the scan/OCR model — not the capture method — is the primary friction driver. The overall upward trend is an artifact of the Android deterioration (I1); iOS and web capture modes remain stable.

NEXT STEP

A live view over git history + .sql / .metrics.json / .insights.json + context_docs/log.md: **7 schemas · 16 tables · 9 MVs · 16 schema commits · 8 context versions** — each commit tied to a context version, diff, and confidence-scored insights.

WHY WE WIN

Built for the unseen spec — not the five we knew.

- **General by design** — parses free-form specs, no hardcoding for known specs.
- **End-to-end & traced** — spec → schema → context → insight in one Langfuse chain.
- **Full stack** — Vector ingestion, ClickStack observability, live schema & funnel dashboards.
- **Insights, not charts** — confidence-scored actions a PM ships today.